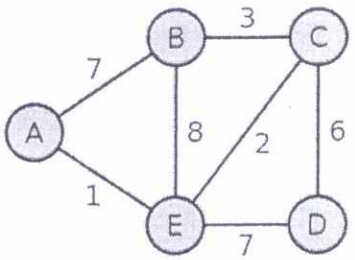
Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
Auxiliary Examination (2024)

Date: 13th August 2024 | Course Code: UCS414/UCS520
Duration: 3 Hours | Course Name: Computer Networks
Max Marks: 100 | Name of Faculty: Dr. Sumit Kumar

Note: Attempt all questions. Assume missing data, if any, suitably.

Q1	(A) Differentiate between Local Area Networks (LANs) and Wide Area Networks (WANs) with suitable examples. (B) Elaborate three responsibilities of the Presentation Layer of the OSI model.	(5) (5)
Q2	Draw and explain the following network topologies with suitable examples and also write their advantages and disadvantages: (A) Ring topology (B) Star Topology	(10)
Q3	(A) Draw the IEEE 802.11 frame format. Explain the sub-fields of the Frame Control (FC) field of the header. (B) Discuss the looping problem in bridge connectivity with a suitable example. Also, demonstrate the solution for the looping problem.	(5) (5)
Q4	Assume that the previous signal level was positive, then draw the signal pattern for the data 100110 using the below-mentioned coding techniques: i) Unipolar NRZ ii) MLT-3 iii) Pseudoternary iv) 2B1Q	(10)
Q5	(A) Assume that in a Stop-and-Wait automatic repeat request (ARQ) system, the bandwidth of the link is 2 Mbps, and 1 bit takes 10 milliseconds to make a round trip. Calculate the bandwidth-delay product. If the system data frames are 1000 bits in length, then compute the utilization percentage of the link. (B) The following decimal numbers represent the 4-bit segments of a data message: 11, 6, 9, and 3. Compute the checksum value for the data message.	(5) (5)
Q6	(A) Explain the working of the "p-persistent" CSMA persistence method with a suitable example. (B) Illustrate the use of "Header Length" and "Protocol" fields in the IPv4 header format with a suitable example.	(5) (5)
Q7	An organization is granted the block 101.17.100.0/24. The administrator wants to create 32 subnets of equal length in the block. Compute the following: (A) Find the subnet mask. (B) Find the number of addresses in each subnet. (C) Find the first and last addresses in the first subnet.	(10)

	(D) Find the first and last addresses in the last subnet.	
Q8	(A) Elaborate the significance of the following ICMP error-reporting messages: i) Source Quench ii) Redirection	(5)
	(B) Illustrate the significance of following closed-loop congestion control mechanisms with suitable examples: i) Backpressure ii) Explicit Signaling	(5)
Q9	(A) Five channels are multiplexed using TDM. If each channel sends 1000 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.	(5)
	(B) A network is modeled as a weighted graph, as shown below. Each router (represented as vertices A, B, C, D, and E) chooses a route to the destination router so that the total cost for the route is the least cost among all possible routes. If the link state routing protocol is implemented on routers, prepare the link state packet for each router. Also, create the Link State Database for router "A" when it receives the link state packets from every other router in the network.  	(5)
Q10	(A) Draw the TCP Segment Format. Discuss the significance of Window size and urgent pointer fields in this segment format.	(5)
	(B) Define the following terms used in Stream Control Transmission Protocol: i) Transmission sequence number ii) Stream sequence number	(5)

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